

Hazidien Ramadhan Utomo

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Summary

GIS and Spatial Analyst with hands-on experience in renewable energy, LiDAR-based spatial analysis, remote sensing, and urban energy assessment. Experienced in ArcGIS and QGIS for spatial modelling, site suitability assessment, spatial data integration, and geospatial visualization. Applied LiDAR-derived nDSM and solar radiation modelling to assess rooftop photovoltaic potential in Surabaya, supported by SolarGIS validation. Also experienced in Google Earth Engine, land-use modelling, and spatial analysis for environmental and infrastructure planning.

Education

Institut Teknologi Sepuluh Nopember

Surabaya, Indonesia

Major in Urban and Regional Planning,

Aug 2022 – Aug 2026

GPA: 3.36/4.00

Achievement:

- 3rd Place Winner of the Infographic Research Competition – GIS THEMATICAL COMPETITION National Level
- 1st place in Football regional competition of Surabaya city Trofeo UINSA

Work Experiences

Assistant Lecturer – Planning Analysis Methods / Metode Analisis Perencanaan

Aug 2025 - Dec 2025

Asisstant Lecturer

Surabaya, Indonesia

Academic teaching support role focused on qualitative, quantitative, and GIS-based spatial planning analysis, supporting laboratory sessions, applied coursework, and student assessment in urban and regional planning

- **Integrated AI-assisted learning tools to support analytical reasoning, logical validation, and structured brainstorming**, guiding students in evaluating qualitative, quantitative, and GIS-based spatial analyses for decision-making relevance.
- Guided spatial reasoning exercises involving **spatial regression, clustering, and spatial autocorrelation (Moran's I)** to assess causal relationships and spatial patterns.
- Designed and reviewed **case-based analytical tasks**, identifying logical gaps, methodological bias, and uncertainty in spatial analysis outcomes.

PT Greenova Daya Prima

Jul 2025 - Aug 2025

Spatial Analysis Assistant – Renewable Energy (Internship)

Surabaya, Indonesia

Energy consulting firm engaged in building energy baseline studies, rooftop solar potential mapping, and spatial analysis to support urban energy transition and decarbonization planning.

- Assessed rooftop solar energy potential using GIS and LiDAR-derived data, integrating **multiple spatial drivers and assumptions under data uncertainty**.
- **Participated in Focus Group Discussions (FGD) with Surabaya City Government and building stakeholders**, supporting coordination and technical clarification during the SETI Surabaya baseline study.
- **Assisted coordination and validation of survey data covering 233 residential and non-residential buildings**, including electricity consumption and building typology classification.

British Embassy Jakarta

Jun 2024 - Nov 2024

Surveyor MRT Surabaya Project

Surabaya, Indonesia

Urban transport planning project focused on transportation demand assessment, traffic movement analysis, and commuter behavior studies to support the planning and development of the Surabaya MRT system.

- Analyzed transportation demand and mobility patterns to **evaluate spatial accessibility, network performance, and system-level implications**.
- Supported validation of traffic performance indicators by **cross-checking survey results, traffic counts, and observed mobility behavior**.
- Contributed to **scenario-based transport planning analysis** under data and operational constraints.

Skills

Geospatial Reasoning & Evaluation:

- Spatial reasoning, causal analysis, scenario-based assessment, uncertainty awareness, model evaluation

GIS & Spatial Modeling:

- ArcGIS, QGIS, Google Earth Engine, spatial statistics, spatial regression, land-use modeling (GeoSOS-FLUS)

Decision Support & Forecasting:

- MCDA (AHP, Delphi), spatio-temporal modeling, environmental & infrastructure planning

Soft Skills:

- Communication, teamwork, analytical problem-solving, Leadership

Project Experiences

Undergraduate Thesis Research

Sep 2025 – Jul 2026

- Designed a **GIS-based spatial analysis methodology** to assess **rooftop solar energy potential at the district level** in Surabaya by integrating building data, land use, and solar radiation parameters.
- Developed a **GIS-based spatial analysis framework** integrating building footprints, land use, LiDAR-derived rooftop data, and solar radiation to assess rooftop PV potential across **31 districts in Surabaya**.
- Conducted **LiDAR-based Area Solar Radiation modelling** in ArcGIS and validated the results against SolarGIS, achieving an **RMSE of 143.40 kWh/m²/year (7.36% relative error)**.

Spatio-Temporal Deforestation Modeling Project – West Sumatra

Nov 2025 – Dec 2025

- **Developed spatio-temporal deforestation analysis workflows on Google Earth Engine (GEE) using AI-assisted code generation**, leveraging multi-temporal Landsat imagery (2009–2024) for large-scale environmental assessment.
- **Critically evaluated and validated AI-generated GEE scripts**, assessing methodological logic, parameter choices, spatial bias, and uncertainty by comparing outputs against observed historical deforestation patterns.
- **Refined AI-assisted analytical workflows to improve reliability and interpretability**, translating validated results into risk-informed insights for environmental monitoring and spatial forecasting under data uncertainty

Baseline Energy Consumption in the Building Sector in Kota Surabaya

Jul 2025 – Aug 2025

- Assisted in compiling and analyzing **baseline electricity consumption data** for residential, commercial, and government buildings in Surabaya.
- Supported **spatial integration of building datasets**, land use data, and energy indicators to enable city-scale energy baseline assessment.
- Contributed to **energy planning analysis** to support urban decarbonization strategies and renewable energy integration.

Rokan Hulu Land Use Modelling Project

Jan 2025 – Feb 2025

- Conducted land use change modeling using GeoSOS-FLUS based on Rokan Hulu land use data (2018 & 2023) to **train and validate ANN-based transition potentials**, and critically interpreted simulation results to assess **causal relationships, scenario plausibility, and model limitations** for long-term spatial forecasting.
- Simulated **land use scenarios for 2023–2043** by integrating driving factors and **Markov Chain** analysis to project future land use dynamics.
- Evaluated model performance using **Kappa statistics and Figure of Merit (FoM)** to assess prediction accuracy and spatial agreement.

MRT Surabaya Project – Transportation Survey & Demand Analysis

Jun 2024 – Nov 2024

- Conducted **transportation demand surveys** (household, employee, and administrative) to analyze commuting behavior and travel patterns.
- Performed **traffic counting and vehicle speed surveys** to assess traffic flow, congestion, and roadway performance.
- Supported **data collection and preliminary analysis** for MRT planning and sustainable urban transport development.

Certification

- **Aksara Lab Indonesia:** Intensive Bootcamp – LiDAR Data Classification & Contour Generation (2025)
- **Geoaccess Indonesia:** Thematic GIS Training – Land Cover Change Prediction using QGIS & GeoSOS-FLUS (2025)
- **Geoaccess Indonesia:** Thematic GIS Training – Flood Modeling Based on Land Use Change & Prediction (2024)
- **Geoaccess Indonesia:** Basic & Thematic Google Earth Engine (GEE) Training – Deforestation Impact on Air Pollution Analysis (2024)
- **Aptanata Community:** Geographic Information System Training – Facility Coverage Enhancement & Road Loading Impact Scenarios (2024)
- **Aptanata Community:** Geographic Information System Training – Service Area & Facility Service Reach Analysis (2024)

Organizational Experiences

UKM Sepakbola ITS

Jan 2024 - Dec 2024

Head of the Football Division

- Organized and developed UKM Sepakbola ITS members, including training coordination and player development programs.
- Coordinated team delegation for regional football championships and served as acting coach during competitions and training sessions when the main coach was unavailable.

Planopolis ITS

Jul 2023 - Dec 2023

Asisstant Manager of Competition

- Led the **planning, organization, and execution** of all competition-related activities for Planopolis ITS, ensuring smooth, timely, and well-coordinated events.
- Developed and implemented the **competition's strategic vision**, aligning programs with the overall objectives of Planopolis ITS and the Department of Urban and Regional Planning.

Aksi Sepuluh Nopember 2023

Aug 2023 - Dec 2023

Vice Chairman I – Campus Orientation Event

- Coordinated and supervised **internal divisions** (events, equipment, logistics, and operations).
- Led **planning, team coordination, and decision-making**, assisting the Event Chairman in execution.
- Successfully delivered a large-scale orientation event that received **media coverage by JTV**, enhancing public visibility and institutional outreach.